



CrossEarth-Gate: Fisher-Guided Adaptive Tuning Engine for Efficient Adaptation of Cross-Domain Remote Sensing Semantic Segmentation

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Abstract

In Remote Sensing (RS), Parameter-Efficient Fine-Tuning (PEFT) has emerged as a key approach to activate the generalizable representation ability of foundation models for downstream tasks. However, existing specialized PEFT methods often fail when applied to large-scale Earth observation tasks, as they are unable to fully handle the multifaceted and unpredictable domain gaps (e.g., spatial, semantic, and frequency shifts) inherent in RS data. To overcome this, we propose CrossEarth-Gate, which introduces two primary contributions. First, we establish a comprehensive RS module toolbox to address multifaceted domain gaps, comprising spatial, semantic, and frequency modules. Second, we develop a Fisher-guided adaptive selection mechanism that operates on this toolbox. This selection is guided by Fisher Information to quantify each module's importance by measuring its contribution to the task-specific gradient flow. It dynamically activates only the most critical modules at the appropriate layers, guiding the gradient flow to maximize adaptation effectiveness and efficiency. Comprehensive experiments validate the efficacy and generalizability of our method, where CrossEarth-Gate achieves state-of-the-art performance on 16 out of 18 cross-domain benchmarks for RS semantic segmentation.

1. Introduction

With the rapid expansion of Geospatial Foundation Models (GFM) in both capability and scale [9, 20, 48, 56], a central challenge emerges: how to efficiently activate their downstream potential. Parameter-Efficient Fine-Tuning (PEFT)

[74, 81] methods aim to match or surpass full fine-tuning while updating only a small fraction of parameters, offering an attractive balance between performance and efficiency. However, when applied to large-scale and globally distributed Earth observation tasks, existing PEFT methods often suffer substantial performance degradation due to highly variable and unpredictable domain gaps [33, 40, 43, 68].

Domain gaps in Remote Sensing (RS) are multifaceted, arising from diverse sources like wavelength ranges, geographical landscapes, and climatic zones. These gaps manifest as a complex interplay of adaptation challenges: (1) **Spatial shifts**, representing the change in object structure and scale that require geometric integrity; (2) **Semantic shifts**, denoting the differences in class appearance and concepts; and (3) **Frequency shifts**, involving high-frequency spectral artifacts or textural noise from different features. However, as shown in Fig. 1 (a), existing PEFTs, whether general-purpose or RS-specific, typically specialize in one functional pathway. For example, LoRA [29] modifies the Multi-Head Self-Attention (MSA) layers to enhance spatial dependency modeling. AdaptFormer [5] adjusts the Multi-Layer Perception (MLP) layers to refine high-level semantic features. The Earth-Adapter [33] focuses on mitigating high-amplitude artifacts in the frequency domain.

While each method has its merits, this single-pathway design is inherently limited: it captures only one facet of RS domain shifts while leaving others unaddressed. As shown by the segmentation maps in Fig. 1(a), LoRA misclassifies large areas of water with high-frequency waves as forest, revealing its inability to process unfamiliar frequency artifacts. AdaptFormer fails to maintain spatial continuity of the road, shattering its structure and demonstrating a critical failure in spatial awareness. Earth-Adapter falters on semantic ambiguities and misidentifies coastal waves as for-

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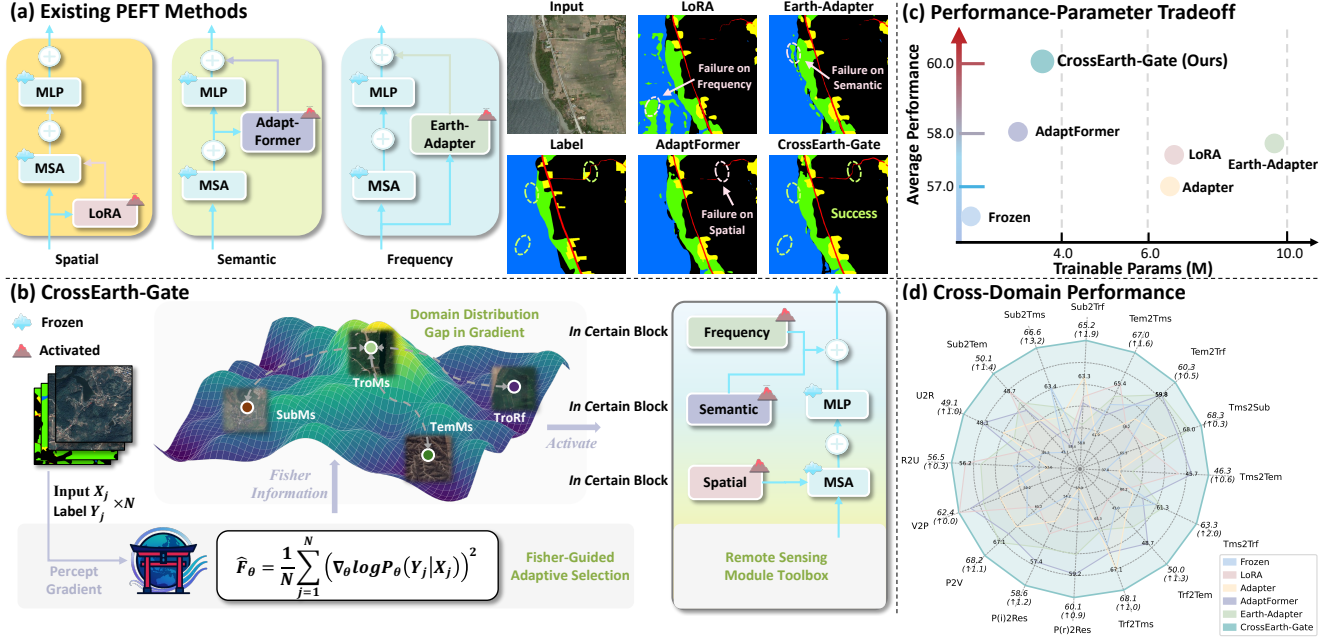


Figure 1. Overview of the CrossEarth-Gate and its comparative advantages. (a) Existing PEFTs typically focus on one specific functional pathway (e.g., LoRA for spatial, Adaptformer for semantic, Earth-Adapter for frequency). The qualitative example of generalization across different climate zones shows each baseline failing on challenges outside its specialty, while our method succeeds. (b) Our proposed CrossEarth-Gate establishes a toolbox combining all three module types. Then, we utilize Fisher Information to guide the gradient flow to activate only the most critical modules at the most relevant blocks for a specific domain. (c) CrossEarth-Gate results in a superior Performance-Parameter Tradeoff. (d) CrossEarth-Gate achieves state-of-the-art results across 16 challenging DG and DA benchmarks.

est, committing a semantic error. These dilemmas motivate us to pursue a more dynamic mechanism capable of jointly addressing spatial, semantic, and frequency variations.

To address this, we propose **CrossEarth-Gate**, a Fisher-Guided adaptive tuning engine for Cross-domain PEFT in Earth observation. Our methods provide a dynamic and comprehensive solution to multifaceted domain gaps, which comprises two components. First, CrossEarth-Gate establishes a structured **RS module toolbox**, designed to inject trainable parameters into a different functional pathway of models, including spatial, semantic, and frequency modules using different PEFT methods. This provides a unified framework with the full capacity to address complex domain shifts. Another novel and critical design is the **Fisher-guided adaptive selection** that operates on this toolbox to dynamically choose the most critical modules at the appropriate layers. Specifically, Fisher Information guides the selection [15], which serves as a metric to evaluate the importance of each module in terms of its contribution to the current task-specific gradient flow. This mechanism periodically determines where to “cast its hook”, directing the gradient flow of the model’s parameters in a way that maximizes adaptation efficiency and effectiveness for domain-specific shifts. This process facilitates a multi-stage learning process, allowing for a robust and nuanced adaptation.

In addition, we rigorously evaluate CrossEarth-Gate

across 18 Domain Generalization (DG) and Domain Adaptation (DA) benchmarks for RS semantic segmentation. It achieves State-Of-The-Art (SOTA) results on 16 of them, outperforming PEFT baselines by up to 3.2% mIoU while offering a better performance-efficiency tradeoff (See Fig. 1 (c) and (d)). The main contributions of this work are:

- We address the challenges posed by extreme heterogeneity and multifaceted domain shifts inherent in RS adaptation tasks with a unified, comprehensive framework.
- We propose CrossEarth-Gate, which introduces a structured RS module toolbox to comprehensively tackle the multifaceted RS domain shifts and a Fisher-guided adaptive selection mechanism to dynamically guide the gradient flow to maximize efficiency and effectiveness.
- CrossEarth-Gate outperforms existing specialized cross-domain methods, FMs, and PEFE approaches across 16 cross-domain benchmarks for RS semantic segmentation, validating its strong generalizability and effectiveness.

2. Related Works

2.1. Geospatial Foundation Models

The concept of Foundation Model (FM) originates from the NLP [72] domain, defined as “the base models trained on large-scale unlabeled data that can be adapted for a variety of downstream tasks” [3, 8, 12, 53, 54, 63, 65, 70]. Driven

by the success of self-supervised learning (SSL) techniques such as masked patch reconstruction, Vision Foundation Models (VFM) have yielded significant achievements in Computer Vision (CV), serving as powerful backbones for general visual feature representation [6, 14, 23, 24, 35, 50].

Adapting existing VFM architectures and techniques to Earth observation data, typically represented as images, has given rise to Geospatial Foundation Models (GFMs) [9, 19, 20, 38, 39, 45, 48, 56, 57, 61, 67, 69, 83]. For example, SatMAE [9] introduces temporal and multi-spectral positional encodings for satellite imagery within the mask image modeling pre-training. To tackle domain shift, CrossEarth [19] utilizes Earth-style injection and multi-task training for domain generalization. However, as the representation capabilities and scale of these FMs increase, how to efficiently and effectively leverage their potential for diverse RS downstream tasks remains a key problem.

2.2. Cross-Domain PEFT in RS

PEFT has emerged as a promising solution to efficiently fine-tune FMs while using a minimum number of trainable parameters [21, 26, 29, 34, 81]. These methods can be broadly classified into four categories: Selective, Additive, Prompt, and Reparameterization approaches [81]. Selective PEFT focuses on only optimizing a specific subset of the model's parameters [2, 4, 60, 75, 80, 85, 86]. For instance, [60] selects sparse parameters based on Fisher information as a measure of parameter importance. Additive PEFT introduces extra trainable parameters into the frozen backbone [5, 26, 52, 52, 78]. Prompt PEFT incorporates learnable prompts into the input or the attention layers to adapt to a specific task. Reparameterization PEFT reformulates or decomposes existing model parameters so that only a subset requires adjustment during fine-tuning [29, 42]. Particularly, LoRA [29] decomposes the updated weight into two low-rank matrices, inspiring numerous subsequent methods [11, 22, 44, 82, 87]. Moreover, existing cross-domain studies begin to explore PEFT for Domain Adaptation (DA) [16, 18, 59, 76] and Domain Generalization (DG) [30, 71].

Recently, PEFT techniques have also gained increasing attention within the RS community, which commonly leverages VFMs or GFMs for various downstream tasks [13, 31, 32, 58, 62, 79, 84]. For instance, SLR [58] and DEFLECT [81] utilize low-rank matrices to adapt to different modalities of RS data, while Earth-Adapter [33] is designed to mitigate the domain shift resulting from frequency-domain artifacts. However, these PEFT methods typically focus on one aspect of the problem, such as spatial, semantic, and frequency adaptation, failing to address the complex, multifaceted nature of RS domain shifts. A method specialized for one aspect often fails when confronted with challenges from another. Consequently, a single static method fails to address these intertwined chal-

lenges comprehensively. To address this, we establish a structured toolbox combining three module types, targeting different functional pathways of the model. We then propose a Fisher-guided adaptive selection mechanism that dynamically identifies the most relevant modules, enabling a more comprehensive cross-domain adaptation.

3. Preliminary

Optimization Objective. Let us consider a cross-domain adaptation task with a training dataset $D = \{(\mathbf{X}, \mathbf{Y})\}$, where the $\mathbf{X} \in \mathbb{R}^{h \times w \times c}$ denotes the input RS images, comprising $h \times w$ pixels with c bands, and $\mathbf{Y}$ represents corresponding label. For Domain Generalization (DG) tasks, D represents the labeled source domain dataset, while Domain Adaptation (DA) additionally includes the unlabeled target domain input and its pseudo-labels. Given a pretrained FM with parameters α , PEFT-related modules with parameters ζ ($|\zeta| \ll |\alpha|$), and the decoder head $\mathcal{H}_\phi$ parameterized by ϕ , the objective of cross-domain PEFT is to enhance the model performance in the target domain by optimizing:

$$\arg \min_{\zeta, \phi} \mathbb{E}_{(\mathbf{X}, \mathbf{Y}) \in D} \mathcal{L}(\mathcal{H}_\phi(\mathcal{B}_{\alpha, \zeta}(\mathbf{X})), \mathbf{Y}), \quad (1)$$

where $\mathcal{B}_{\alpha, \zeta}$ signifies the backbone incorporating the PEFT modules and $\mathcal{L}$ is the loss function for the adaptation task.

Transformer Block. This work mainly focuses on Transformer-based [65] FMs, such as ViT [14]. The input $\mathbf{X}$ is first divided into l patches and then encoded into d -dimensional embedding through a patch embedding layer, yielding the initial tokens $\mathbf{T}_1 \in \mathbb{R}^{l \times d}$. Subsequently, $\mathbf{T}_1$ is processed by a sequence of I transformer blocks in the FM. We denote the input features of i -th block as $\mathbf{T}_i \in \mathbb{R}^{l \times d}$. Each block is typically composed of a Multi-head Self-Attention (MSA) module and a feed-forward Multi-Layer Perceptron (MLP) network, with residual connections and Layer Normalization (LN) applied after each module. The computation for the i -th block is defined as follows:

$$\mathbf{T}_i^{\text{attn}} = \text{MSA}(\mathbf{T}_i) + \mathbf{T}_i, \quad \mathbf{T}_i^{\text{attn}} \in \mathbb{R}^{l \times d}, \quad (2)$$

$$\mathbf{T}_{i+1} = \text{MLP}(\mathbf{T}_i^{\text{attn}}) + \mathbf{T}_i^{\text{attn}}, \quad \mathbf{T}_{i+1} \in \mathbb{R}^{l \times d}, \quad (3)$$

where we omit normalization for simplicity and $\mathbf{T}_{i+1}$ is the output of i -th block, which is passed as input to next block.

4. Methods

The Fig. 1 (b) presents an overview of the CrossEarth-Gate, which consists of a structured RS module toolbox and Fisher-guided adaptive selection mechanism.

4.1. Remote Sensing Module Toolbox

First, to equip the model with the comprehensive capabilities needed to tackle intertwined domain gaps, CrossEarth-Gate first establishes a structured and comprehensive RS

module toolbox. This toolbox integrates spatial, semantic, and frequency modules, each engineered into distinct functional pathways to provide specific aspects of the feature hierarchy for specific domain challenges. Instead of statically choosing one method for certain layers, we initially insert all these modules at every layer of the pre-trained backbone.

Spatial Module. The MSA sub-layer (Eq. (2)) explicitly models the relationships between tokens, allowing the model to build contextual understanding and capture spatial dependencies at various scales [25, 47]. In the RS application scenarios, adapting to change in object scale, spatial arrangement, or geographic layout necessitates a targeted tuning of this reasoning mechanism. Therefore, we employ the Spatial Module using Low-Rank Adaptation (LoRA) [29]. LoRA is predicated on the hypothesis that the weight update $\Delta\mathbf{W}$ for a pre-trained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times d}$ during adaptation possesses a low “intrinsic rank”. This update can be parameterized by the product of two low-rank matrices, $\mathbf{A} \in \mathbb{R}^{d \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d}$, where $r \ll d$ is the rank. The merged weight $\mathbf{W}$ is written as:

$$\mathbf{W} = \mathbf{W}_0 + \Delta\mathbf{W} = \mathbf{W}_0 + \mathbf{B}\mathbf{A}, \quad \mathbf{W} \in \mathbb{R}^{d \times d}. \quad (4)$$

We strategically inject these trainable, low-rank matrices into the query ($\mathbf{W}_Q$) and value ($\mathbf{W}_V$) linear projection weights of each MSA module (Eq. (2)). This targeted intervention directly modulates the self-attention mechanism, influencing how the model weighs and aggregates spatial information. It allows the model to adapt its understanding of spatial context and object scales without altering the parameters in the original, frozen $\mathbf{W}_Q$ and $\mathbf{W}_V$ weights.

Semantic Module. The MLP sub-layer (Eq. (3)) transforms each token’s representation independently, which is widely understood to be a key locus of factual and semantic knowledge within the model [17, 49, 77]. Moreover, generalizing concepts across different domains (e.g., “building” in rural and urban landscapes) logically requires modifying this stored knowledge. To adapt the model’s high-level semantic knowledge, we introduce a Semantic Module based on the Adapter architecture [5, 26], placed in parallel with each MLP sub-layer. The adapter in i -th block comprises a down-projection layer with weight $\mathbf{W}_i^{down} \in \mathbb{R}^{d \times \hat{d}}$, a GELU activation, and an up-projection layer with weight $\mathbf{W}_i^{up} \in \mathbb{R}^{\hat{d} \times d}$, satisfying $\hat{d} \ll d$. The output of this module is added to the original MLP’s output via a residual connection. Given the modified output $\hat{\mathbf{T}}_i^{attn}$ of the MSA integrating LoRA, the modified output of block $\hat{\mathbf{T}}_{i+1}$ becomes:

$$\hat{\mathbf{T}}_{i+1} = \text{MLP}(\hat{\mathbf{T}}_i^{attn}) + \text{Adapter}_i(\hat{\mathbf{T}}_i^{attn}) + \hat{\mathbf{T}}_i^{attn}, \quad (5)$$

$$\text{Adapter}_i(\hat{\mathbf{T}}_i^{attn}) = \text{GELU}(\hat{\mathbf{T}}_i^{attn} \cdot \mathbf{W}_i^{down}) \cdot \mathbf{W}_i^{up}. \quad (6)$$

This parallel structure allows the module to refine or adjust the semantic transformation performed by the frozen MLP without disrupting the original pre-trained knowledge flow.

Frequency Module. RS imagery is uniquely challenged by high-amplitude artifact influences, which are almost situated everywhere in the RS image. To address this, we introduce Frequency Modules based on the Earth-Adapter [33], which operates in the frequency domain. The module first utilizes the Fourier Transform to decompose the input features into low-frequency (structural) and high-frequency (detail/texture) components. These disentangled components are processed by distinct, lightweight adapter experts. A mixture-of-adapters router then learns to selectively process and recombine these frequency components, effectively mitigating artifact disturbances while preserving essential features. Given the modified output of the i -th block $\hat{\mathbf{T}}_{i+1}$, the input to next block $\hat{\mathbf{T}}_{i+1}$ becomes:

$$\hat{\mathbf{T}}_{i+1} = \hat{\mathbf{T}}_{i+1} + \text{Earth-Adapter}_i(\hat{\mathbf{T}}_{i+1}). \quad (7)$$

4.2. Fisher-Guided Adaptive Selection

Given the toolbox, the central challenge becomes determining which modules to activate for a specific task. Tuning all modules simultaneously is fundamentally inefficient, yet a static or heuristic selection would fail to efficiently leverage the full potential of the toolbox for diverse domain shifts, reverting to a sub-optimal strategy. Therefore, CrossEarthGate employs a principled, data-driven selection mechanism as the dynamic guide. We conceptualize the adaptation as a gradient flow: as the model fine-tunes, task-specific gradients flow through the network, and our different module types offer distinct pathways for this flow. Our goal is to periodically analyze this flow and “gate” it, directing it only to the modules that offer the highest impact for adaptation.

To quantify this impact in a theoretically-grounded way, we turn to the Fisher Information Matrix (FIM) [15]. A parameter’s significance can be determined by evaluating the extent to which altering it influences the model’s output. Given a model parameterized by $\theta \in \mathbb{R}^{|\theta|}$, we denote the output distribution of the model as $P_\theta(\mathbf{Y}|\mathbf{X})$ for input $\mathbf{X}$. Subsequently, we can assess how much a small parameter perturbation $\delta \in \mathbb{R}^{|\theta|}$ in the parameter changes the distribution using the Kullback-Leibler divergence [37] $D_{KL}(P_\theta(\mathbf{Y}|\mathbf{X}) \parallel P_{\theta+\delta}(\mathbf{Y}|\mathbf{X}))$. As shown in [1, 51], when $\delta \rightarrow 0$, the following second-order approximation holds:

$$\mathbb{E}_{\mathbf{X}} [D_{KL}(P_\theta(\mathbf{Y}|\mathbf{X}) \parallel P_{\theta+\delta}(\mathbf{Y}|\mathbf{X}))] = \delta^T F_\theta \delta + O(\delta^3), \quad (8)$$

where $F_\theta \in \mathbb{R}^{|\theta| \times |\theta|}$ is the FIM [15], defined as:

$$F_\theta = \mathbb{E}_{\mathbf{X}} [\mathbb{E}_{\mathbf{Y} \sim P_\theta(\mathbf{Y}|\mathbf{X})} \nabla_\theta \log P_\theta(\mathbf{Y}|\mathbf{X}) \nabla_\theta \log P_\theta(\mathbf{Y}|\mathbf{X})^T]. \quad (9)$$

Apparently, the FIM links δ to the resultant changes in the model’s output distribution. However, the $|\theta| \times |\theta|$ size of F_θ renders its exact computation infeasible. Consequently, we adopt the empirical diagonal approximation of the FIM [36, 60], which is equivalently a vector in $\mathbb{R}^{|\theta|}$. When sampling N data pairs $(\mathbf{X}_j, \mathbf{Y}_j)$ from the task dataset D , the diagonal FIM for θ can be empirically estimated as:

$$\hat{F}_\theta = \frac{1}{N} \sum_{j=1}^N (\nabla_\theta \log P_\theta(\mathbf{Y}_j | \mathbf{X}_j))^2, \quad (10)$$

where $\log P_\theta(\mathbf{Y}_j | \mathbf{X}_j)$ is realized as the negative task loss. This practical approximation relates the $\hat{F}_\theta$ to the square gradient of the parameter. We can conceptualize the adaptation as a flow, where the task-specific loss directs gradients back to the parameters. The gradient represents the direction and magnitude of this flow. A large $\hat{F}_\theta$ value signifies that this parameter is a “high-flow” channel, *i.e.*, a pathway where gradients are large and thus highly impactful on the model’s output. This is precisely where we should “cast our hook” to achieve the maximum adaptation gain.

The FIM approximation in Eq. (10) provides an importance score for an individual parameter. To select entire modules, CrossEarth-Gate employs a dynamic gating mechanism that periodically re-evaluates module importance throughout the fine-tuning process. By using the empirical FIM as our “flow meter”, we are able to dynamically “gate” the gradient flow, directing it only to the modules that offer the highest impact. Specifically, at every N training iterations, the framework will temporarily activate all modules in the toolbox. The Fisher information is then computed by summing over a small batch of M data samples. To derive a module-level importance score S_i^z for the z -th type of module parameterized by ζ_i^z in the i -th block, we aggregate the scores of all its constituent parameters: $\hat{S}_i^z = \sum_{\zeta_i^z} \hat{F}_{\zeta_i^z}$. To make the scores of different types of modules comparable, we compute the category-normalized relative importance scores: $S_i^z = \frac{\hat{S}_i^z}{\sum_i \hat{S}_i^z}$, where I represents the number of blocks. This ensures that the selection process is balanced, making all module types comparable and promoting a diverse, task-specific configuration. Subsequently, only Top- k modules with the highest scores are activated, and the gradient flow is gated to only these pathways for the next N training iterations. CrossEarth-Gate repeats this process to dynamically adapt its tuning strategy to the most critical components for the task at hand. The hyperparameter configurations are listed in the Appendix.

5. Experiments

5.1. Settings

We conduct a comprehensive evaluation of CrossEarth-Gate on DG and DA benchmarks for RS semantic segmentation

tasks. Additional dataset details, implementation specifics, and baseline configurations are provided in the Appendix.

Datasets. Our cross-domain analysis leverages a suite of diverse datasets. The CASID dataset specifically addresses domain shifts across four climate zones: Subtropical Monsoon (Sub), Temperate Monsoon (Tem), Tropical Monsoon (Tms), and Tropical Rainforest (Trf). The ISPRS Potsdam and Vaihingen contain aerial images collected over the cities of Potsdam and Vaihingen, with IR-R-G, R-G-B, and R-G-B-IR channels. RescueNet consists of aerial imagery focused on detecting buildings impacted by disasters to facilitate rescue operations. LoveDA presents a cross-scene dataset with RS images from both urban and rural areas. For DG experiments, we follow the protocol from [33]. We employ CASID to establish 12 out-of-domain generalization scenarios. We also evaluate on RescueNet with source domain as the RGB (P(r)2Res) and IR–R–G (P(I)2Res) channels of Potsdam. For DA experiments, we form four benchmarks: Potsdam to Vaihingen (P2V), Vaihingen to Potsdam (V2P), Rural to Urban (R2U), and Urban to Rural (U2R). These datasets provide a spanning multifaceted RS domain shifts, including unseen geographical regions, novel spectral bands, and diverse climatic conditions.

Implementation Details. All experiments are implemented based on the MMSegmentation [10] framework. We mainly employ Dinov2 [50] as the feature extraction backbone, paired with Mask2Former [7] as the segmentation head. For DA experiments, we utilize the DACS [64] self-training framework, whereas for DG experiments, we train the model via end-to-end supervised learning. The models are finetuned using AdamW [46] optimizer with a base learning rate of 1e-5 for the decoder and PEFT modules.

Baselines. We compare CrossEarth-Gate against existing PEFT methods, including VPT [34], SLR [58], Rein [71], LoRA [29], Adapter [26], Adaptformer [5], and Earth-Adapter [33]. We also select two conventional fine-tuning approaches: “Frozen”, where all backbone parameters are fixed, and “Full-Tuning”, where we fine-tune the entire model. Additionally, for the DG experiments, we include results from two cross-domain specialized models and the Full-Tuning of different GFMs, based on [19].

5.2. Cross-Domain Performance Comparison

Generalization Across Climate Zones. As presented in Tab. 1, Full-Tuning undergoes a catastrophic performance decline compared to the Frozen backbone, confirming that unconstrained fine-tuning severely overfits to the source domain. Specialized, static PEFTs present inconsistent performance, showing varying efficacy depending on the domain. For instance, the Earth-Adapter achieves the second-

Table 1. Performance comparison on CASID benchmarks across 12 DG experiments. We compare specialized DA models, FMs, PEFT methods, and CrossEarth-Gate. The **best** and second-best scores are indicated in bold and underlined, and the improvement of CrossEarth-Gate is shown in brackets. Sub: Subtropical Monsoon. Tem: Temperate Monsoon. Tms: Tropical Monsoon. Trf: Tropical Rainforest.

Backbone	Method	Params (M)	Sub2Tem	Sub2Tms	Sub2Trf	Average	Tem2Sub	Tem2Tms	Tem2Trf	Average
MiT-B5 [73]	HRDA [28]	81.4	34.6	63.0	56.2	51.3	62.7	63.3	44.9	57.0
	DAFormer [27]	81.4	36.0	63.5	59.0	52.8	59.8	63.8	51.7	58.4
ViT-L [14]	MTP [66]	304.2	39.4	60.0	61.6	53.7	66.2	64.7	48.3	59.7
	SatMAE [9]	304.2	33.1	52.9	48.7	44.9	51.8	43.7	36.2	43.9
	ScaleMAE [56]	304.2	39.2	60.7	56.4	52.1	62.5	57.3	45.9	55.2
	RemoteCLIP [41]	304.2	8.5	15.1	19.7	14.4	54.3	57.8	16.7	42.9
DINOv2-L [50]	Frozen	0.0	43.1	63.4	58.8	55.2	66.9	64.8	59.0	63.6
	Full-Tuning	304.2	38.6	62.3	58.3	53.0	58.7	55.8	43.8	52.8
	CrossEarth [19]	3.0	48.1	64.6	64.2	59.0	63.5	61.8	57.8	61.0
	VPT [34]	3.1	41.2	61.4	60.8	54.4	64.6	61.2	58.7	61.5
	SLR [58]	6.9	24.3	37.0	32.3	31.2	33.3	22.0	26.1	27.1
	Rein [71]	3.0	42.3	56.2	52.9	50.5	66.0	62.1	56.7	61.6
	LoRA [29]	6.4	48.7	60.5	62.3	57.2	68.2	65.4	58.2	64.0
	Adapter [26]	6.3	46.9	61.5	63.3	57.2	67.7	61.9	58.5	62.7
	Adaptformer [5]	3.2	47.9	58.4	62.1	56.1	68.2	64.4	59.8	64.1
	Earth-Adapter [33]	9.6	48.3	61.6	60.5	56.8	68.8	64.9	59.8	64.5
	CrossEarth-Gate	3.0-4.4	50.1 (+1.4)	66.6 (+2.0)	65.2 (+1.0)	60.6 (+1.6)	68.0 (-0.8)	67.0 (+1.6)	60.3 (+0.5)	65.1 (+0.6)
Backbone	Method	Params (M)	Tms2Sub	Tms2Tem	Tms2Trf	Average	Trf2Sub	Trf2Tem	Trf2Tms	Average
MiT-B5 [73]	HRDA [28]	81.4	63.8	33.3	56.9	51.3	63.3	43.0	63.1	56.5
	DAFormer [27]	81.4	61.8	31.4	56.2	49.8	62.9	39.6	62.7	55.1
ViT-L [14]	MTP [66]	304.2	56.1	32.3	60.1	49.5	55.6	37.0	59.8	50.8
	SatMAE [9]	304.2	60.2	32.5	53.0	48.6	59.1	32.8	56.7	49.5
	ScaleMAE [56]	304.2	62.1	35.2	55.4	50.9	60.6	30.9	58.0	49.8
	RemoteCLIP [41]	304.2	22.5	12.3	19.1	19.1	32.1	26.0	23.6	27.2
DINOv2-L [50]	Frozen	0.0	65.3	37.0	60.9	54.4	68.0	43.0	66.7	59.2
	Full-Tuning	304.2	60.9	29.9	58.9	49.9	61.9	32.9	64.3	53.0
	CrossEarth [19]	3.0	69.1	40.5	60.7	56.8	67.9	42.3	64.4	58.2
	VPT [34]	3.1	64.8	33.9	56.8	51.8	67.3	38.0	66.9	57.4
	SLR [58]	6.9	39.6	23.6	38.7	34.0	42.2	24.2	34.3	33.5
	Rein [71]	3.0	58.2	29.0	53.4	46.9	62.6	33.8	61.1	52.5
	LoRA [29]	6.4	66.7	45.2	60.2	54.5	67.7	46.4	65.3	59.8
	Adapter [26]	6.3	67.8	38.5	60.4	55.5	68.9	47.7	67.1	61.2
	Adaptformer [5]	3.2	67.9	45.7	60.5	58.0	69.8	48.7	66.8	61.8
	Earth-Adapter [33]	9.6	68.0	39.2	61.3	56.2	68.4	47.8	65.7	60.6
	CrossEarth-Gate	3.0-4.4	68.3 (+0.3)	46.3 (+0.6)	63.3 (+2.0)	59.3 (+1.3)	69.0 (-0.8)	50.0 (+1.3)	68.1 (+1.0)	62.4 (+0.6)

best average result (64.5%) when trained on the Tem domain, whereas the Adaptformer performs better (61.8%) when trained on the Trf domain. This variance suggests that different climate-zone transfers necessitate a different emphasis on spatial, semantic, or frequency adjustments, exposing the inherent limitations of the specialized PEFT strategy. In contrast, CrossEarth-Gate demonstrates consistent performance, achieving SOTA results in 10 of the 12 scenarios and securing the highest average mIoU across all four source domains, all while maintaining high parameter efficiency. This superior result highlights the critical advantage of our structured RS module toolbox, which comprehensively tackles the complex RS domain shifts. While our method is dominant, in two isolated cases (Tem2Sub and Trf2Sub), static methods like Earth-Adapter or Adaptformer achieve marginally better results. This suggests that for these specific shifts, their fixed module placement coincidentally aligns well with the required feature adaptation. However, these static, specialized methods fail to generalize to other domains. The comprehensive results underscore the robustness and superior generalization capability of the CrossEarth-Gate framework across different climate zones.

Generalization to Disaster Scenarios. Tab. 2 shows the results of the generalization task from a standard aerial dataset (Potsdam) to a post-disaster scenario (RescueNet), involving simultaneous shifts in geography, spectral bands, and target object appearance. CrossEarth-Gate again achieves the highest mIoU in both experiments, with an mIoU of 60.1% on P(r)2Res and 58.6% on P(i)2Res. Our method demonstrates strong generalization on ambiguous, large-scale classes like Impervious surfaces and Clutter, which are prevalent in post-disaster environments. However, for the Building class, the Frozen backbone outperforms our method in the P(r)2Res case. This is a noteworthy finding, suggesting that the DINOv2’s pre-trained representations for highly structured objects are already extremely robust and generalizable. In this context, any adaptation may marginally disrupt these near-optimal features for the Building class. Despite these class-specific variances, the comprehensive SOTA mIoU scores validate the superior and more balanced generalization capability of CrossEarth-Gate in the complex cross-domain RS scenarios.

Adaptation to Unlabeled Target Domains. The DA results presented in Tab. 3, where the model has access to

Table 2. Performance comparison on P(r)2Res and P(i)2Res DG experiments. Surf: Impervious surfaces. Bldg: Building. Clut: Clutter.

Backbone	Method	Params (M)	P(r)2Res (Classes)					mIoU (%)	P(i)2Res (Classes)					mIoU (%)
			Surf	Bldg	Tree	Car	Clut		Surf	Bldg	Tree	Car	Clut	
MiT-B5 [73]	HRDA [28]	81.4	28.8	27.2	56.8	4.0	66.4	36.6	26.1	29.2	44.3	5.2	61.6	33.3
	DAFormer [27]	81.4	33.0	32.6	55.0	7.6	66.7	39.0	34.1	43.6	41.7	12.0	58.9	38.1
ViT-L [14]	MTP [66]	304.2	34.3	40.9	41.7	16.3	62.7	39.2	31.3	36.8	0.5	14.4	57.5	28.1
	SatMAE [9]	304.2	13.7	13.8	26.1	3.7	27.1	16.9	15.5	13.7	24.4	3.0	21.1	15.5
	ScaleMAE [56]	304.2	23.5	31.9	45.4	0.1	47.7	29.7	13.9	35.0	48.7	0.3	49.4	29.4
	RemoteCLIP [41]	304.2	2.2	8.2	2.6	0.0	0.0	2.6	1.3	10.1	27.5	0.2	0.8	8.0
DINOv2-L [50]	Frozen	0.0	45.0	61.7	64.1	33.8	66.3	54.2	53.0	61.4	56.5	31.1	69.1	54.2
	Full-Tuning	304.2	21.3	22.5	25.2	3.0	61.8	26.7	17.5	29.6	64.5	3.2	65.3	36.0
	CrossEarth [19]	3.0	43.6	59.7	51.0	16.4	63.2	46.8	41.9	60.6	51.3	10.6	62.9	45.5
	VPT [34]	3.1	50.3	56.6	58.8	38.1	72.0	55.1	52.8	61.0	58.8	35.6	71.1	55.9
	SLR [58]	6.9	52.1	59.3	61.0	21.7	72.8	53.4	54.4	57.4	55.5	19.3	71.3	51.6
	Rein [71]	3.0	48.8	59.9	51.4	23.0	69.4	50.5	21.1	42.0	56.3	19.9	59.7	39.8
	LoRA [29]	6.4	51.1	57.4	62.5	39.1	72.8	56.6	50.1	59.8	65.1	<u>35.1</u>	72.3	56.5
	Adapter [26]	6.3	56.7	59.6	53.1	30.3	69.8	53.9	<u>56.1</u>	59.3	63.5	<u>30.7</u>	<u>73.3</u>	56.1
	Adaptformer [5]	3.2	<u>58.5</u>	60.3	<u>65.5</u>	38.9	<u>73.1</u>	<u>59.2</u>	55.4	61.9	60.2	37.1	72.6	<u>57.4</u>
	Earth-Adapter [33]	9.6	56.5	<u>61.1</u>	63.4	<u>39.6</u>	72.4	58.6	54.2	60.8	61.8	<u>35.1</u>	72.8	57.0
	CrossEarth-Gate	3.3-4.0	60.8 (+2.3)	58.6 (-3.1)	67.5 (+2.0)	40.5 (+0.9)	73.3 (+0.2)	60.1 (+0.9)	57.7 (+1.6)	<u>61.5</u> (-0.4)	<u>64.6</u> (-0.5)	34.6 (-2.5)	74.7 (+1.4)	58.6 (+1.2)

Table 3. Performance comparison on DA benchmarks between representative DA models, PEFT methods, and CrossEarth-Gate.

Methods	Params (M)	Domain Adaptation				Avg.
		P2V	V2P	R2U	U2R	
HRDA [28]	81.4	67.6	58.6	53.2	35.3	53.7
DAFormer [27]	81.4	64.4	54.8	52.7	42.5	53.6
Frozen	0.0	66.1	59.2	54.9	45.5	56.4
Full-Tuning	304.2	62.4	59.6	42.6	35.8	50.1
VPT [34]	3.1	65.0	57.6	55.8	47.0	56.4
SLR [58]	6.9	14.4	16.5	23.8	9.6	16.1
Rein [71]	3.0	64.1	59.5	54.7	34.5	53.2
LoRA [29]	6.4	65.2	62.4	<u>56.2</u>	46.6	57.6
Adapter [26]	6.3	66.8	60.3	53.6	47.7	57.1
Adaptformer [5]	3.2	66.9	62.2	53.7	<u>48.1</u>	57.7
Earth-Adapter [33]	1.0-3.9	<u>67.1</u>	61.6	56.0	47.5	<u>58.1</u>
CrossEarth-Gate	1.7-3.9	68.2 (+1.1)	62.4 (+0.0)	56.5 (+0.3)	49.1 (+1.0)	59.1 (+1.0)

unlabeled target data, further validate our approach. Full-Tuning fails again, likely due to instability from noisy target-domain pseudo-labels. In contrast, PEFT methods, including LoRA, Adaptformer, and Earth-Adapter, all outperform the strong Frozen baseline, demonstrating their benefits in the DA context. However, CrossEarth-Gate achieves the best performance across four adaptation scenarios with the highest average mIoU of 59.1%. This consistent success highlights the efficacy of our Fisher-guided selection in the DA context. The gradient flow, informed by pseudo-labels from a specific target domain, allows our model to precisely identify and activate only the most critical modules needed for that particular adaptation. While our method tied with the LoRA on the V2P task, this does not indicate a limitation. Rather, it suggests that this specific V2P adaptation may be predominantly a spatial-scale challenge, which LoRA’s static architecture coincidentally addresses well. However, CrossEarth-Gate’s ability to win or tie across all diverse UDA scenarios validates its more

robust and principled approach to dynamic adaptation.

5.3. Ablation Studies

Backbone Generalizability. As detailed in Tab. 4, we evaluate CrossEarth-Gate across five pre-trained VFMs and GFMs on the CASID benchmark, including DINOv2 series [50], SAM [35], SatMAE [9], and Scale-MAE [56]. Specifically, Full-Tuning is an unstable and generally poor strategy for DG. It performs significantly worse than the Frozen baseline on the DINOv2 series, a clear case of overfitting to the source domain, while incurring an enormous parameter cost. In contrast, CrossEarth-Gate consistently and significantly outperforms both the Frozen and Full-Tuning baselines across all five architectures. It achieves the highest average mIoU in every case, with improvements from 2.1 % to 7.6 % mIoU, while tuning a minimal number of parameters. This robust performance strongly validates the remarkable versatility and generalizability of the CrossEarth-Gate.

Core Components Impact. We further dissect the framework by ablating its key components, with results visualized in Fig. 3. When all modules from the toolbox are trained simultaneously without selection, it results in the most performance degradation. Its failure proves that naively increasing trainable parameters is detrimental, likely leading to conflicting gradient updates. This result empirically validates the necessity of CrossEarth-Gate to guide the gradient flow for utilizing adaptation resources effectively. Removing each specialized module type causes a comparable drop in mean performance. All three adaptation pathways are non-redundant and essential for achieving comprehensive generalization. For instance, the Sub and Tem domains suffer most from the removal of spatial and frequency modules, while Trf is most impacted by the loss of semantic modules, underscoring the varied nature of these domain challenges.

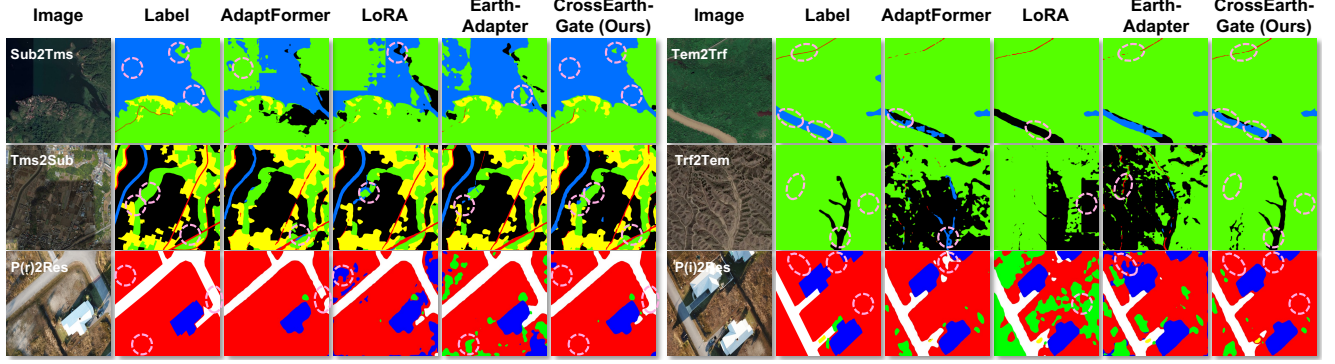


Figure 2. Visualizations of predicted segmentation maps of PEFT methods. In the CASID [43] dataset, red is the road class, yellow is the building class, blue is the water class, green is the forest class, and black is the background class. In the RescueNet [55] dataset, white is the impervious surface class, red is the clutter class, blue is the building class, green is the vegetation class, and yellow is the car class.

Table 4. Ablation studies of model backbone on CASID benchmarks. We only show the average performance of four climate domains as the source domain, respectively. We demonstrate the generalizability of CrossEarth-Gate across different backbones.

Backbone	Method	Params (M)	Source Domain				Mean
			Sub	Tem	Tms	Trf	
SatMAE (Large) [9]	Frozen	0.0	24.4	21.5	24.8	26.3	24.3
	Full-Tuning	304.2	35.4	14.9	35.2	29.8	28.8
	CrossEarth-Gate	2.9-4.0	37.1	21.8	36.8	31.8	31.9
Scale-MAE (Large) [56]	Frozen	0.0	50.7	48.5	50.9	50.9	50.3
	Full-Tuning	304.2	52.8	57.6	51.9	52.6	53.7
	CrossEarth-Gate	2.9-3.6	56.3	58.1	54.9	53.8	55.8
SAM (Huge) [35]	Frozen	0.0	53.3	57.5	53.2	53.4	54.4
	Full-Tuning	631.2	54.7	58.2	51.7	54.6	54.8
	CrossEarth-Gate	4.0-4.9	56.1	61.9	56.3	59.3	58.4
DINOv2 (Small) [50]	Frozen	0.0	49.5	59.6	51.9	52.7	53.4
	Full-Tuning	22.1	49.4	49.3	48.7	50.2	49.4
	CrossEarth-Gate	0.7	54.6	62.0	52.7	55.8	56.2
DINOv2 (Base) [50]	Frozen	0.0	51.3	62.5	52.8	54.4	55.3
	Full-Tuning	86.6	50.2	53.7	50.2	49.4	50.9
	CrossEarth-Gate	1.4-1.8	55.6	63.6	54.7	57.4	57.8

5.4. Qualitative Comparison

We also provide a qualitative comparison in Fig. 2, visualizing the predictions from our method, LoRA, AdaptFormer, and Earth-Adapter across six DG scenarios. The PEFT baselines, while effective for their intended challenge, fail outside their specialization. For example, the semantic-focused AdaptFormer fails to detect large areas of water with frequency artifacts in the Sub2Tms task. The spatial-related LoRA misclassifies the forest as water in the Tms2Sub task, resulting in semantic errors. The frequency-focused Earth-Adapter demonstrates spatial discontinuity of the road in the Tem2Trf task. Notably, CrossEarth-Gate avoids the catastrophic semantic errors, correctly captures the spatial extent of objects, and produces clean, accurate maps, mitigating the influence of artifacts. Our method con-

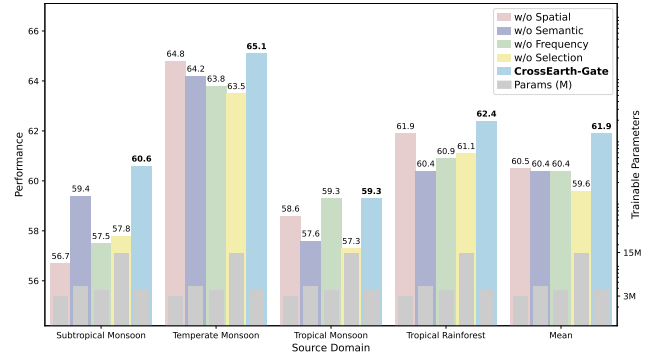


Figure 3. Ablation study of model component on CASID benchmarks. We compare the performance and trainable parameters of CrossEarth-Gate against versions with key components removed.

sistently produces segmentation maps that are closer to the ground-truth labels across all six scenarios, demonstrating a more robust and superior generalization capability.

6. Conclusion

This paper addresses the limitations of current static, specialized PEFT methods in cross-domain RS adaptation. Existing approaches often fall short by focusing on a single facet of the complex spatial, semantic, and frequency shifts inherent to RS data. We introduce CrossEarth-Gate to tackle this limitation through two key innovations: a comprehensive RS module toolbox and a Fisher-guided adaptive selection mechanism. The toolbox integrates three module types to provide distinct functional pathways of models. Subsequently, by leveraging Fisher Information to quantify module importance, CrossEarth-Gate dynamically directs the gradient flow, activating only the most critical modules for the specific adaptation task. Comprehensive experiments demonstrate the effectiveness, efficiency, generalizability, and explainability of our method.

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